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Beyond Problem Solving: A Three-Space Probe of LLM Critical Thinking

 (/pdf?id=Bce9yI6zGd)

Author names redacted to protect student authors.

 29 Jul 2025 (modified: 31 Aug 2025)  ACL ARR 2025 July Submission

 July, Senior Area Chairs, Area Chairs, Reviewers, Authors  Revisions (/revisions?id=Bce9yI6zGd)

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TL;DR:  We propose a new important metric: critical thinking evaluation the LLM performance on inconsistent questions

Abstract:

Large language models (LLMs) are widely used for complex reasoning tasks, with various datasets designed to evaluate their problem-solving capabilities across different domains. However, limited attention has been given to LLM performance on unsolvable questions caused by inconsistencies. This paper proposes an evaluation framework grounded in Hierarchical Three-Space Theory to assess LLMs' critical thinking capabilities, specifically their ability to identify problem inconsistencies and challenge flawed problem setups. We design two types of inconsistency: insufficient information that renders answers indeterminate, and redundant information that introduces contradictions. We apply these modifications to multiple-choice problems spanning different fields and generative math problems. We establish detailed behavioral categories for LLM responses and analyze whether LLMs recognize inconsistencies and how they respond to them. Our experiments with state-of-the-art LLMs reveal significant limitations in critical thinking, demonstrating that critical thinking represents a distinct capability that cannot be improved solely through enhanced reasoning or problem-solving skills. Through comprehensive experiments, we analyze how prompting strategies and problem difficulty influence critical thinking performance.

Paper Type: Long

Research Area: Resources and Evaluation

Research Area Keywords: benchmarking, evaluation, metrics

Contribution Types: Model analysis & interpretability

Languages Studied: English

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

Preprint:  no

Preprint Status:  There is no non-anonymous preprint and we do not intend to release one. (this option is binding)

Preferred Venue:  AACL

Consent To Share Data:  yes

Consent To Share Submission Details:  On behalf of all authors, we agree to the terms above to share our submission details.

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: No

A2 Elaboration: The paper propose a new evaluation metric: critical thinking of LLMs, and evaluate the model performance on question-answering task. All the datasets are widely used math or science questions, there is no pontential risk

B Use Or Create Scientific Artifacts: Yes

B1 Cite Creators Of Artifacts: Yes

B1 Elaboration: We introduce the datasets and models in Sec 3.2

B2 Discuss The License For Artifacts: N/A

B3 Artifact Use Consistent With Intended Use: Yes

B3 Elaboration: Sec 1, Sec 3

B4 Data Contains Personally Identifying Info Or Offensive Content: No

B4 Elaboration: The datasets are widely used question-answer datasets, includin science and math problems

B5 Documentation Of Artifacts: Yes

B5 Elaboration: Sec 1, Sec 3

B6 Statistics For Data: Yes

B6 Elaboration: Sec 3, Sec 4

C Computational Experiments: Yes

C1 Model Size And Budget: Yes

C1 Elaboration: Sec 3.2

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: Sec 3

C3 Descriptive Statistics: Yes

C3 Elaboration: Sec 4

C4 Parameters For Packages: N/A

D Human Subjects Including Annotators: Yes

D1 Instructions Given To Participants: Yes

D1 Elaboration: Sec 3.2, Appendix B

D2 Recruitment And Payment: No

D2 Elaboration: The research group members create the curated small scale validation dataset

D3 Data Consent: Yes

D3 Elaboration: Sec 3.2

D4 Ethics Review Board Approval: Yes

D4 Elaboration: Sec 3

D5 Characteristics Of Annotators: Yes

D5 Elaboration: Sec 3

E Ai Assistants In Research Or Writing: Yes

E1 Information About Use Of Ai Assistants: Yes

E1 Elaboration: Sec 3.2

Author Submission Checklist: yes

Association For Computational Linguistics - Blind Submission License Agreement:  On behalf of all authors, I agree

Submission Number: 1122

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Meta Review of Submission1122 by Area Chair P5pv

Meta Review by Area Chair P5pv 📅 24 Sept 2025, 05:42 (modified: 02 Oct 2025, 12:04)

👁 Senior Area Chairs, Area Chairs, Authors, Reviewers Submitted, Program Chairs

📄 Revisions (/revisions?id=HGU2VDE4Zj)

Metareview:

This paper investigates the critical thinking ability of large language models (LLMs), distinguishing it from problem-solving. Based on the Three-Space Theory—problem framing, strategy, and implementation—the authors construct an evaluation framework that rewrites existing tasks into unsolvable variants by removing essential information or introducing contradictions. Applied to multiple-choice and generative problems across mathematics, logic, and medicine, the framework evaluates models such as GPT-4o and Llama3.3-70B using new metrics: applied and inherent critical thinking rates. Results show that LLMs struggle to detect flawed problems, with critical thinking only weakly correlated with problem-solving ability.

Summary Of Reasons To Publish:

The idea of separating critical thinking from problem-solving in LLMs is somewhat novel.

This paper proposes a framework grounded in the Three-Space Theory, shifting evaluation toward judging problem validity rather than only solving problems.

This paper provides detailed experimental data and specifics, making results reproducible and reliable.

Summary Of Suggested Revisions:

The Three-Space Theory is prominently introduced but not tightly integrated with the experimental design.

The study equates critical thinking solely with detecting deliberately introduced inconsistencies.

Using GPT-4o to evaluate other models' responses risks bias, as judgments may reflect GPT-4o's internal criteria rather than an objective ground truth

Overall Assessment: 2.5 = Borderline Findings

Reported Issues: 👁 No

Publication Ethics Policy Compliance: I used a privacy-preserving tool exclusively for the use case(s) approved by PEC policy, such as language edits

Add: **Author-Editor Confidential Comment**

Official Review of Submission1122 by Reviewer hrVg

Official Review by Reviewer hrVg 📅 01 Sept 2025, 09:54 (modified: 09 Sept 2025, 11:34)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hrVg

📄 Revisions (/revisions?id=96z2jmbxG8)

Paper Summary:

This paper proposes that an LLM's critical thinking ability is different from its problem-solving ability, and advocates evaluating a model's critical thinking under the three spaces of "problem framing space, strategy space, and implementation space." The authors systematically rewrite existing data to construct two types of inconsistencies: 1) Insufficient information: deleting the correct option and hiding necessary conditions; 2) Redundant/misleading: adding gaslight hints or contradictory conditions; and they evaluate the model's critical thinking ability from three dimensions: problem-solving rate, applied critical thinking rate, and inherent critical-thinking rate.

Summary Of Strengths:

The concept that the authors brought about is interesting at a glance.

The authors provide detailed experimental data and specifics, making it easy to reproduce the results.

Summary Of Weaknesses:

1. The authors' "three-space theory" appears to have little connection with the actual experimental design: although the authors devote a large portion to introducing the three-space theory, the specific metrics in the experimental design are not tightly associated with or matched to the three spaces. The three-space theory appears flashy but insubstantial, and does not truly provide better modeling or understanding of the model's critical thinking ability.

2. The definition of critical thinking is too narrow, being defined only as the model's ability to identify (deliberately set) errors in the instruction. However, critical thinking should also have a broader meaning, such as recognizing both the strengths and weaknesses of something. Judging the relationship between critical thinking and problem-solving ability solely through the model's ability to identify errors in the instruction is one-sided and arbitrary.

Comments Suggestions And Typos:

Please further clarify the connection between evaluating the model's critical thinking ability and the three spaces, or better demonstrate the model's specific performance across the three spaces. More comprehensively and accurately model the model's critical thinking ability.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 1 = No usable software released.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

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by Authors (👁)

id=aclweb.org/ACL/ARR/2025/July/Submission1122/Authors))

📅 15 Sept 2025, 23:13

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer hrVg

Comment:

Thank you for your constructive suggestions. Hope these clarifications resolve the questions.

Weakness:

1. The authors' "three-space theory" appears to have little connection with the actual experimental design: although the authors devote a large portion to introducing the three-space theory, the specific metrics in the experimental design are not tightly associated with or matched to the three spaces. The three-space theory appears flashy but insubstantial, and does not truly provide better modeling or understanding of the model's critical thinking ability.

Author Response: We design distinct behavioral patterns to characterize model responses in question-answer tasks and employ these behavioral categories to evaluate the model's critical thinking capabilities. The challenge ratio and F1 score are calculated based on these categories. Detailed descriptions designed for multiple-choice and math questions are provided in Appendix B. The behavioral categories include descriptions of different problem-solving processes related to three-space theory. In future work, we will provide a more detailed analysis of these behavioral patterns to demonstrate a better understanding of the model's critical thinking capability levels.

Furthermore, the basic hypotheses that problem difficulty and solving strategy influence the model's critical thinking are grounded in the three-space theory framework. We assume the LLM problem-solving process follows the three-space framework, whose solving process is formulated as an interaction between different spaces. The model space represents the model's assumptions and understanding about the question when only the question content is provided. During the problem-solving process, LLMs can apply different rules to solve the problem based on different pre-trained knowledge. Different rules pose potential inconsistencies with varying levels of obviousness, thus influencing the final LLM decision: to challenge or believe the completeness of the question context.

2. The definition of critical thinking is too narrow, being defined only as the model's ability to identify (deliberately set) errors in the instruction. However, critical thinking should also have a broader meaning, such as recognizing both the strengths and weaknesses of something. Judging the relationship between critical thinking and problem-solving ability solely through the model's ability to identify errors in the instruction is one-sided and arbitrary.

Author Response: We acknowledge that critical thinking could play a more important and versatile role in the cognitive problem-solving process. In this work, we focus on whether the LLM considers alternative answers that are not provided by the user. We also measure whether the model's problem-solving process is limited by inconsistencies. The critical thinking capability is decomposed into two stages: identifying the inconsistency and recognizing that the correct solution should not be influenced by this inconsistency. Sometimes, even when the inconsistency is detected, LLMs can still provide wrong or unreliable answers. The different behavioral patterns in the reasoning process are described in the categories outlined in Appendix B. Using only one score to represent the entire process could be insufficient; in future work, we will present an analysis of the two stages separately.

Add: **Author-Editor Confidential Comment**

Official Review of Submission1122 by Reviewer XThf

Official Review by Reviewer XThf 📅 31 Aug 2025, 10:53 (modified: 09 Sept 2025, 11:34)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer XThf

📄 Revisions (/revisions?id=jIROv5gqjh)

Paper Summary:

This paper examines whether LLMs can recognize and challenge flawed problems and proposes an evaluation framework based on the Three Space Theory, which includes problem understanding, strategy, and execution. It turns normal problems into two unsolvable types by removing essential information or injecting internal contradictions. The approach is applied to multiple choice and generative items in mathematics, medicine, and logic, and model responses are labeled into fine grained behavioral categories to quantify error recognition and response. The authors define metrics separate from base accuracy, namely applied critical thinking rate and inherent critical thinking rate, and analyze how prompt instructions, reasoning strategies such as Chain of Thought, and problem difficulty and the clarity of inconsistencies affect performance. They conclude that improving accuracy alone does not yield critical thinking and argue that data, prompts, and training should explicitly target problem validity judgment as a separate learning and evaluation objective.

Summary Of Strengths:

1. Define a method for judging problem validity using the Three-Space Theory.

2. Systematically transform answerable items into two types—insufficient information and contradictory information—so it can be applied across domains and formats.
3. Identify the correlation between accuracy and critical thinking, and summarize how prompts, CoT, and problem difficulty affect critical thinking and performance to provide practical guidance.

Summary Of Weaknesses:

1. Ambiguity of the “critical thinking” construct: The paper defines critical thinking based on a three-stage theory, but it is unclear how the actual metrics (Applied Critical Thinking Rate, Inherent Critical-Thinking Rate) reflect this theory. Calculating an F1 score using true positives and false positives is close to a conventional classification evaluation. Can this really accurately represent the complex cognitive ability of “critical thinking”? Although the paper defines critical thinking as a complex process, in practice it measures a very simple outcome. The F1 score only checks whether the AI ultimately classified a problem as “ill-posed,” and it does not present evidence of what internal reasoning the AI followed to reach that conclusion.
2. Looking at Appendix Prompt A, the method of generating data with insufficient information and contradictions may have led LLMs to learn “data-generation rules” rather than “critique.” This may mean the AI did not recognize logical inconsistencies in the problem but simply adapted to abnormal data patterns.
3. The approach of having GPT-4o label other AI models’ answers makes it difficult to ensure objectivity and reliability. GPT-4o may already possess a high level of critical-thinking capability on its own. Therefore, GPT-4o ends up judging whether other models’ answers are “critical” according to its own internal criteria. While labeling can be delegated to GPT, I think this is insufficient as decisive evidence to prove the paper’s central claim that “critical thinking” is a real, independent capability. Human evaluation also appears necessary.
4. The experiment analyzing the effect of “reasoning mode” is limited to the Qwen3 model, making it difficult to generalize the claim to other LLMs.
5. The paper’s claims are interesting, but the structure is scattered and the logical flow is unclear, making it difficult for readers to grasp the core content. The authors need to refocus the paper and connect each section more logically to improve overall readability.

Comments Suggestions And Typos:

1. The authors need to analyze the observed 'critical behavior' more deeply, moving beyond a simple binary success/failure. For instance, a more granular analysis should investigate at which specific reasoning stage the failure occurred—did the model detect the inconsistency but fail to propose a correct solution, or did it fail to detect the inconsistency altogether? This level of detailed analysis would provide a much deeper insight into each model's critical thinking capabilities.
2. A core contribution of this paper is the creation of 'flawed' problems by modifying existing benchmarks. However, without the public release of this dataset, the results remain irreproducible. The authors must upload the entire modified benchmark dataset to a public repository (e.g., Hugging Face Datasets..) to allow other researchers to replicate the experiment under identical conditions and verify the results.
3. The paper uses GPT-4o as the primary labeling tool, while human evaluation is relegated to a supplementary role. However, judging a complex cognitive ability like 'critical thinking' remains a subjective and nuanced task that should not be left entirely to an AI. Therefore, the GPT-generated labels should be used only as a reference, while at least two independent human evaluators perform multiple annotations on each model's response to ensure objectivity and reliability.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 2 = Poor: Some of the main claims are not sufficiently supported. There are major technical/methodological problems.

Excitement: 2 = Potentially Interesting: this paper does not resonate with me, but it might with others in the *ACL community.

Overall Assessment: 2 = Resubmit next cycle: I think this paper needs substantial revisions that can be completed by the next ARR cycle.

Limitations And Societal Impact:

None

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 2 = They would be hard pressed to reproduce the results: The contribution depends on data that are simply not available outside the author's institution or consortium and/or not enough details are provided.

Datasets: 2 = Documentary: The new datasets will be useful to study or replicate the reported research, although for other purposes they may have limited interest or limited usability. (Still a positive rating)

Software: 2 = Documentary: The new software will be useful to study or replicate the reported research, although for other purposes it may have limited interest or limited usability. (Still a positive rating)

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

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by Authors (👁️), +1 more (/group/edit?id=aclweb.org/ACL/ARR/2025/July/Submission1122/Authors))
📅 15 Sept 2025, 23:14
👁️ Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer XThf

Comment:
Thank you for your insightful feedback. In the response, we will express the concept with more details and hope the response alleviates your concerns.

Weakness:

1. Ambiguity of the “critical thinking” construct: The paper defines critical thinking based on a three-stage theory, but it is unclear how the actual metrics (Applied Critical Thinking Rate, Inherent Critical-Thinking Rate) reflect this theory. Calculating an F1 score using true positives and false positives is close to a conventional classification evaluation. Can this really accurately represent the complex cognitive ability of “critical thinking”? Although the paper defines critical thinking as a complex process, in practice it measures a very simple outcome. The F1 score only checks whether the AI ultimately classified a problem as “ill-posed,” and it does not present evidence of what internal reasoning the AI followed to reach that conclusion.

Author Response: We design distinct behavioral patterns to characterize model responses in question-answer tasks and employ these behavioral categories to evaluate the model's critical thinking capabilities. The challenge ratio and F1 score are calculated based on these categories. Detailed descriptions designed for multiple-choice and math questions are provided in Appendix B. The behavioral categories include descriptions of different problem-solving processes related to three-space theory. In future work, we will provide a more detailed analysis of these behavioral patterns to demonstrate a better understanding of the model's critical thinking capability levels.

2. Looking at Appendix Prompt A, the method of generating data with insufficient information and contradictions may have led LLMs to learn “data-generation rules” rather than “critique.” This may mean the AI did not recognize logical inconsistencies in the problem but simply adapted to abnormal data patterns.

Author Response: The instruction is designed to use GPT-4o to generate desired descriptions for the math problem. We believe the current rule is sufficient for the model to generate the desired inconsistent questions and to further verify changes in the number of numerical conditions. Any clarification regarding the necessity of 'critique' would be appreciated.

3. The approach of having GPT-4o label other AI models' answers makes it difficult to ensure objectivity and reliability. GPT-4o may already possess a high level of critical-thinking capability on its own. Therefore, GPT-4o ends up judging whether other models' answers are "critical" according to its own internal criteria. While labeling can be delegated to GPT, I think this is insufficient as decisive evidence to prove the paper's central claim that "critical thinking" is a real, independent capability. Human evaluation also appears necessary.

Author Response: For experiments at this scale, automatic evaluation is necessary. LLM-as-judge methods can be biased and unreliable. However, we do not directly use GPT-4o to judge whether the response is 'critical.' We employ designed behavioral categories to describe different responses and use these categories to determine whether the response demonstrates critical thinking. The detailed criteria and alignment with human annotations are displayed in Appendix B. We have iteratively refined the prompt template and compared it with human labels to achieve the current version.

4. The experiment analyzing the effect of "reasoning mode" is limited to the Qwen3 model, making it difficult to generalize the claim to other LLMs.

Author Response: We agree that investigating multiple reasoning models is important to provide comprehensive results. We also have a reasoning model based on Llama that was fine-tuned through the DeepSeek distillation process. In future work, we could incorporate reasoning models from the GPT series, such as GPT-o1

5. The paper's claims are interesting, but the structure is scattered and the logical flow is unclear, making it difficult for readers to grasp the core content. The authors need to refocus the paper and connect each section more logically to improve overall readability.

Author Response: We will reorganize the results and analysis section to emphasize the main contributions and provide a more logical flow. We aim to highlight the general performance of current LLMs and investigate the two hypotheses derived from the three-space framework: that solving strategy and problem difficulty influence the model's critical thinking capabilities. We have implemented different experimental setups to evaluate various problem-solving strategies and problem difficulty levels.

Add: **Author-Editor Confidential Comment**

Official Review of Submission1122 by Reviewer Jz1i

Official Review by Reviewer Jz1i  31 Aug 2025, 07:49 (modified: 09 Sept 2025, 11:34)

 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Jz1i

 Revisions (/revisions?id=SAcCYBRi5h)

Paper Summary:

This paper focuses on the critical thinking ability of Large Language Models (LLMs) in addressing unsolvable problems containing contradictions. Based on the three-level space theory (Problem Framework Space, Strategy Space, and Implementation Space), it constructs an evaluation framework. Specifically, it modifies multiple-choice questions across various domains and generative mathematics problems through two approaches: removing key information (leading to answer uncertainty) and adding redundant information (causing contradictions). Subsequently, it evaluates the performance of mainstream models such as GPT-4o and Llama3.3-70B. The study reveals that LLMs exhibit significant limitations in critical thinking, and their critical thinking ability only shows a weak positive correlation with problem-solving ability. Additionally, factors like prompting strategies (e.g., the 3-ICL-cha instruction) and problem difficulty impact the performance of critical thinking. Meanwhile, the paper points out that critical thinking is an ability independent of reasoning and problem-solving.

Summary Of Strengths:

1. **Novel Evaluation Framework:** Introduces a novel framework based on the cognitive science "Three-Space Theory" to evaluate LLM critical thinking, shifting the focus from simply solving problems to identifying flaws in the problem's premise.
2. **Systematic Benchmark Creation Methodology:** Provides a general and reproducible method for creating evaluation benchmarks by systematically introducing specific inconsistencies (insufficient or redundant information) into existing, well-defined questions.
3. **Comprehensive Experiments and In-depth Analysis:** Conducts extensive experiments across diverse models and tasks, offering deep insights into how factors like prompting strategies, reasoning capabilities, and problem difficulty influence critical thinking performance.

Summary Of Weaknesses:

1. The study defines critical thinking primarily as the ability to detect inconsistencies (missing or contradictory information) within a self-contained prompt. This overlooks broader, crucial aspects such as evaluating source credibility, identifying logical fallacies, or synthesizing information from multiple conflicting external sources.
2. **Implicit nature of real-world inconsistencies:** In reality, flawed premises in user queries, media bias, or scientific discrepancies are rarely explicit. Instead, they are subtly woven into the context and narrative, without clear signposts like a "Hint." The artificial and direct nature of the contradictions tested here therefore differs significantly from the nuanced challenges models face in authentic scenarios.

Comments Suggestions And Typos:

1. This study focuses on identifying internal inconsistencies within a prompt. Future research should evaluate the detection of external inconsistencies, such as fact-checking claims against outside knowledge, identifying misinformation, or spotting logical fallacies, to better model real-world critical thinking.
2. The study uses static, single-turn tasks. A valuable next step would be to assess critical thinking in multi-turn, interactive settings like dialogues or collaborative problem-solving.

Typos:

1. Line 869 "...cannot be determined due to this inconsistency" -> "...cannot be determined due to this inconsistency"
2. Line 881 (Appendix B, Table 1 Caption): "...the accuracy sis above 90%" -> "...the accuracy is above 90%"

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3.5

Excitement: 3.5

Overall Assessment: 3.5 = Borderline Conference

Ethical Concerns:

There are no concerns with this submission

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 3 = Potentially useful: Someone might find the new datasets useful for their work.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Reviewer Certification: I certify that the review I entered accurately reflects my assessment of the work. If you used any type of automated tool to help you craft your review, I hereby certify that its use was restricted to improving grammar and style, and the substance of the review is either my own work or the work of an acknowledged secondary reviewer.

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📅 15 Sept 2025, 23:14

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Authors, Reviewer Jz1i

Comment:

Thank you for your constructive feedback and suggestions for future directions. Hope the response alleviates your concerns.

Weakness:

1. The study defines critical thinking primarily as the ability to detect inconsistencies (missing or contradictory information) within a self-contained prompt. This overlooks broader, crucial aspects such as evaluating source credibility, identifying logical fallacies, or synthesizing information from multiple conflicting external sources.

Author Response: We agree that a more detailed analysis of reasoning steps will provide greater insights into LLMs' critical thinking capabilities. We have designed different behavioral pattern descriptions in Appendix B to calculate the challenge ratio. These categories represent different behaviors during the reasoning steps, but displaying only the final F1 score may overlook performance discrepancies in the reasoning process. In future work, we will provide more results based on the category distribution and design metrics for the reasoning process, such as checking logical coherence and addressing conflicting information.

Weakness:

2. Implicit nature of real-world inconsistencies: In reality, flawed premises in user queries, media bias, or scientific discrepancies are rarely explicit. Instead, they are subtly woven into the context and narrative, without clear signposts like a "Hint." The artificial and direct nature of the contradictions tested here therefore differs significantly from the nuanced challenges models face in authentic scenarios

Author Response: We acknowledge that inconsistencies can be introduced from various sources. In this work, we start from the most straightforward question-answering scenario with a convenient evaluation metric design. We will explore new datasets or modification methods to create implicit or real-world-like inconsistencies. We also plan to examine more complex scenarios, such as agentic LLMs serving multiturn interactions, including multiturn conversations.

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